Rohan Bhide

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Education

University of California-San Diego (UCSD) — San Diego, CA
Master of Science (MS) - Computer Science; GPA 3.9/4.0 — *Sep 2022 – Dec 2023*

Pune Institute of Computer Technology (PICT) — Pune, India
Bachelor of Engineering (BE) - Computer Engineering; GPA 9.67/10.0 — *Jun 2018 – May 2022*

Experience

Sony Electronics — San Diego, CA
Sr. Software Engineer — *Mar 2024 – Present*

- Optimized sliding window KV cache allocation across QNN, Vulkan, XNNPack using ring buffers, incremental ring masking, and chunked prefill, with 80%+ KV reduction at 4K+ context and 7x faster TTFT via cache rewind.
- Fused the multi-op ring buffer update into a single indexed cache update GLSL shader, eliminating CPU fallbacks and graph fragmentation to achieve 17% faster prefill and 22% faster load in ExecuTorch's Vulkan.
- Achieved 60% size reduction and 92% function-calling accuracy by fine-tuning Gemma/Llama with LoRA and quantizing with mixed-precision PTQ (layer-wise sensitivity analysis, outlier clipping) and QAT (4/8 bit).
- Benchmarked latency, memory, thermals across NPU/GPU/CPU on SM8850, Tensor G4 with per-kernel profiling.
- Expanded 3D ecosystem for Spatial Reality Displays (SR1/SR2) by developing Sony's first C++ OpenXR runtime, enabling support for 20+ professional applications including Omniverse, VRED, Blender, and TwinMotion.
- Ensured sub-16ms motion-to-photon latency by engineering a multi-threaded mirror window for DirectX/OpenGL interop via shared handles and synchronization fences, integrating Nvidia Reflex for performance measurement.
- Enabled high-performance LLM inference on devices with less than 2GB RAM by pruning and distilling Llama3 to sub-1B parameters, facilitating CPU-only execution integrated with Whisper and LangChain for embedded RAG.
- Secured a RAG-based troubleshooting assistant against adversarial threats, maintaining <10% successful attacks on high severity tests by implementing Promptfoo red-teaming and token-optimized prompt engineering.
- Scaled creative insights to 150K+ professionals by architecting a serverless AWS platform and Android app leveraging multimodal Gemini models for real-time composition analysis and on-device framing guidance.

VR Software Engineering Intern — *Oct 2023 – Mar 2024*

- Enabled native, glasses-free 3D previewing within the Maya viewport by developing a C++ plugin that captures dual real-time camera perspectives using off-axis projection for Spatial Reality Displays (SR1/SR2).
- Improved spatial accuracy and framing efficiency for 3D artists by implementing viewport overrides and custom hardware-representative gizmos that constrain camera frustums to the gizmo's physical bounds.
- Ensured seamless UI/UX consistency across 8+ varied unit and scale configurations by creating a MEL-based management interface that automates gizmo lifecycle and alignment logic.
- Reduced deployment overhead by 60% by architecting an Azure DevOps CI/CD pipeline utilizing Visual Studio and WiX Toolset to automate compilation, signed installer generation, and delivery of DLLs, shaders, and scripts.

The Design Lab @UCSD — San Diego, CA
Graduate Researcher — *Oct 2022 – Dec 2023*

- Advanced driving safety research by programmatically generating 26 complex route configurations in Python for DReyeVR (CARLA/Unreal Engine), modeled after federal crash data to simulate high-risk scenarios.
- Engineered high-fidelity multiplayer Unity FPS with NavMesh AI pathfinding and raycast combat, incorporating diegetic UI for emotion visualization to achieve a 150% increase in self-reported social presence scores.
- Quantified impact of affective projection on virtual social presence through a user study (N=18) by integrating AffectNet facial expression recognition via low-latency UDP sockets. [escholarship.org/uc/item/2xv7w1zq]

AutoVRse — Bengaluru, India
AR/VR Software Engineering Intern — *Feb 2022 – Jul 2022*

- Built an industrial assembly simulation using VRTK by leading 3 developers, overseeing the Git workflow, and configuring Unity SmartMerge to resolve complex scene conflicts.
- Reduced enterprise training time by 75% by developing a VR power tool simulation for Meta Quest featuring C# voice-guided lessons and HLSL shaders for pulsing spatial highlights.
- Engineered a cooperative multiplayer skating system via 6-DOF controller velocity translation and hosted 200+ users in VRChat by deploying network-optimized object synchronization for shared interactable items using Udon.

Persistent Systems Pune, India
AR/VR Software Engineering Intern *Jun 2021 – Aug 2021*

- Streamlined consumer networking hardware installation via a Unity/ARFoundation app, using image tracking to overlay dynamic 3D guides for 7 components (incl. LAN, Power, RJ-11) synced with troubleshooting instructions.
- Enabled dynamic instruction delivery and remote feedback management via Firebase Realtime Database REST APIs; improved accessibility by developing text-to-speech, voice commands, and interactive modules.

Computational Linguistics Lab @PICT Pune, India
Undergraduate Researcher *Sep 2020 – Aug 2021*

- Architected end-to-end Tesseract and OpenCV OCR pipeline to digitize 20+ textbooks into structured formats.
- Created a dataset of 4000+ textbook questions and achieved 88% Bloom's Taxonomy classification accuracy by fine-tuning BERT with a multi-label classification head in PyTorch.
- Reduced manual sourcing time by 50% using Matplotlib/Seaborn heatmaps for cognitive complexity distribution.

iMocha Pune, India
Natural Language Processing Intern *Mar 2020 – Sep 2020*

- Demonstrated 81% accuracy evaluating candidate essay grammar, tone, and relevance by fine-tuning BERT in PyTorch and integrating the inference pipeline into a Django backend with robust error-handling.
- Automated analysis of candidate fluency and pronunciation for 1,000+ interviews via Azure's Speech-to-Text APIs.

Projects

XOdia 2019: AI vs AI Competition Platform | *github.com/rhn19/Xodia_2019*

- Enabled execution of C++ bot matches within resource-limited Docker environments using Django with threaded producer-consumer queues and reCAPTCHA-protected user models.
- Optimized search heuristics for sub-second decision-making in Abalone (10^{154} complexity) by engineering C++ agents utilizing MiniMax with Alpha-Beta pruning.

IndicLM: Language Models for Indian Languages | *github.com/rhn19/indic-LM*

- Developed tokenizers for 10+ Indic languages by training BPE, WordPiece, and SentencePiece on 10M+ IndicNLP and Wikipedia samples using PyTorch and Hugging Face Transformers.
- Attained 37.06 perplexity and 96% accuracy on downstream tasks for Marathi by training an ALBERT model via Masked Language Modeling.

TCGuard: Legal Clause Analysis using Deep Learning | *github.com/rhn19/devhack*

- Identified risky EULA clauses with 96% accuracy by training a PyTorch Bi-LSTM on 23,000 legal documents and deploying the model via Flask.

High-Performance Entity Simulation (Unity DOTS/ECS) | *github.com/rhn19/ECS-Zombies*

- Simulated 5,000+ autonomous agents at high frame rates by engineering a Unity ECS system utilizing Burst Compiler, parallel job scheduling, and cache-friendly memory.

Technical Skills

Languages: C++, Python, C#, SQL, Bash, GLSL
Frameworks: PyTorch, ExecuTorch, TorchAO, HuggingFace Transformers, LangChain, OpenXR, Unity, Unreal Engine, DirectX, OpenGL, Vulkan, ARFoundation, Django, Flask, QNN
Tools: Git, Docker, AWS, Azure DevOps, OpenCV, WiX Toolset, Maya, Visual Studio, Android Studio, Xcode, iOS, Android, Windows, Linux, Spatial Reality Display
Expertise: On-Device/Edge AI, LLM Fine-tuning/Quantization/Pruning/Distillation, GenAI (RAG), AR/VR/XR, 3D Math, Multi-threading, NLP